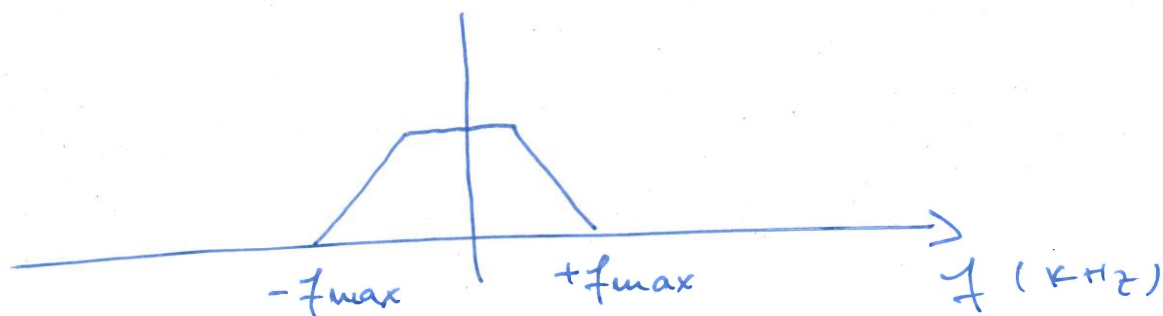


A/D TUTORIAL

- SAMPLING, ALIASING, QUANTISATION
 - ANTI-ALIAS FILTER
 - ANTI-IMAGE FILTER
 - SQNR
 - BANDPASS SAMPLING
 - OVERSAMPLING

Q1

An Analogue signal has the spectrum shown below:



Assume $f_{\max} = 15 \text{ kHz}$

- (1) Determine the min sampling frequency to avoid aliasing.
- (2) Sketch the spectrum of the signal after sampling, indicating the relevant frequencies including the Nyquist frequency if

(i) $F_s = 20 \text{ kHz}$

Sketch the interval $\pm 20 \text{ kHz}$

(ii) $F_s = 30 \text{ kHz}$

Sketch the interval $\pm 30 \text{ kHz}$

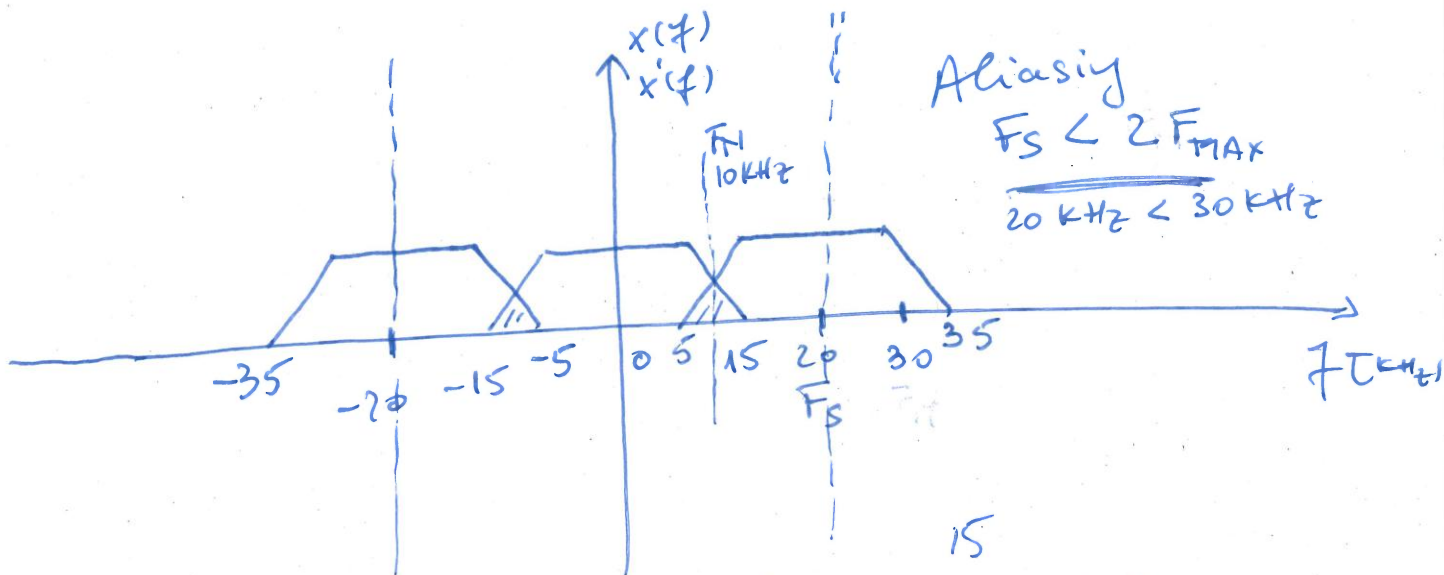
(iii) $F_s = 40 \text{ kHz}$

Sketch the int $\pm 40 \text{ kHz}$

- (3) Compare (i) (ii) (iii) and comment on practical implications

(i) $F_s = 20 \text{ kHz}$

(3a)

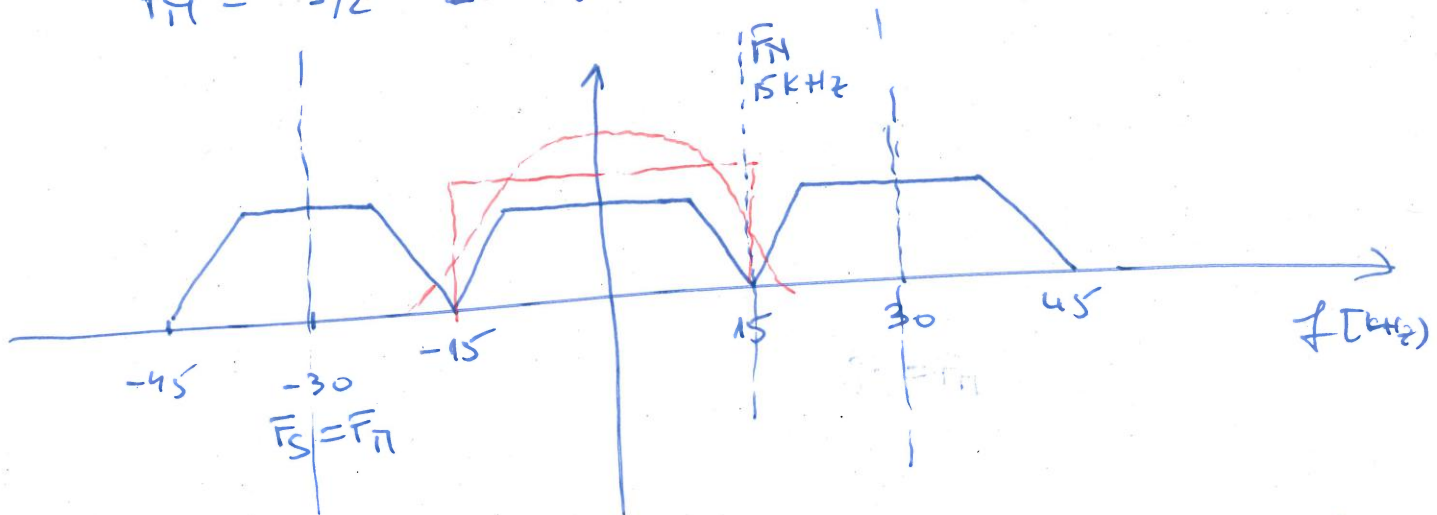


Nyquist frequency $F_N = F_s/2 = 10 \text{ kHz} < F_{MAX}$
Practical incorrect signal

(ii) $F_s = 30 \text{ kHz}$

$F_N = F_s/2 = 15 \text{ kHz} = F_{MAX}$

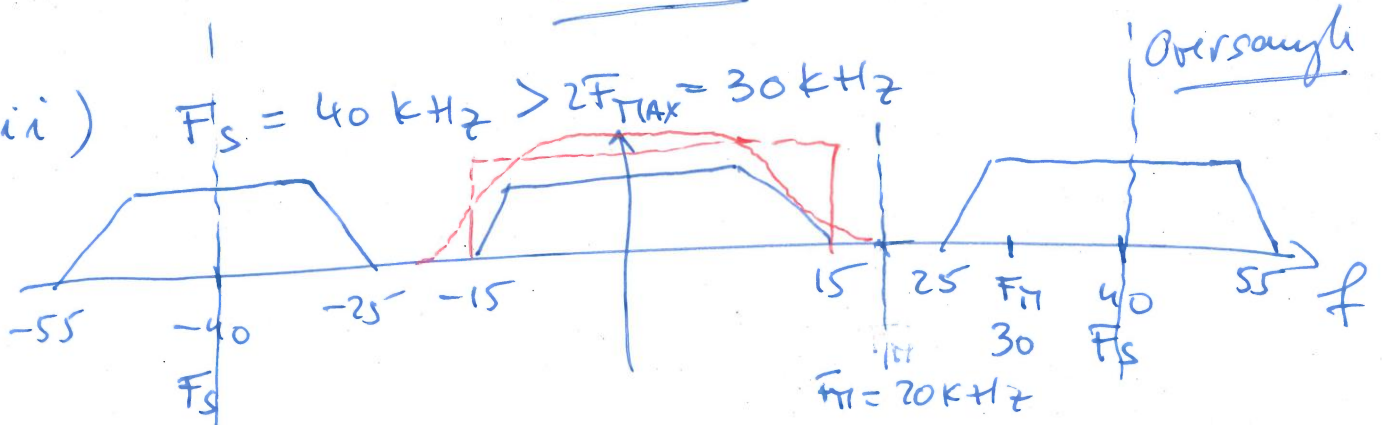
Sampled at
Nyquist
 $F_s = 2F_{MAX}$



Practical minimum complexity

(iii) $F_s = 40 \text{ kHz} > 2F_{MAX} = 30 \text{ kHz}$

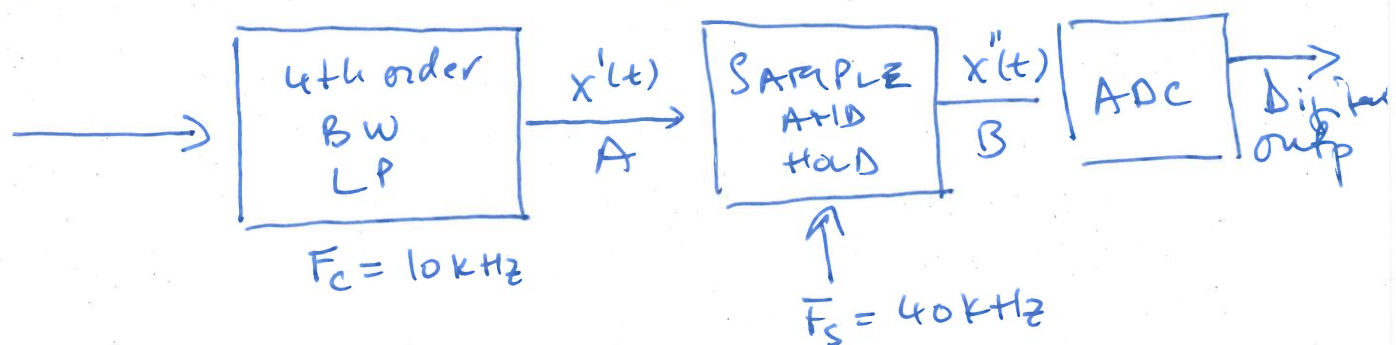
Oversample



Q2

4

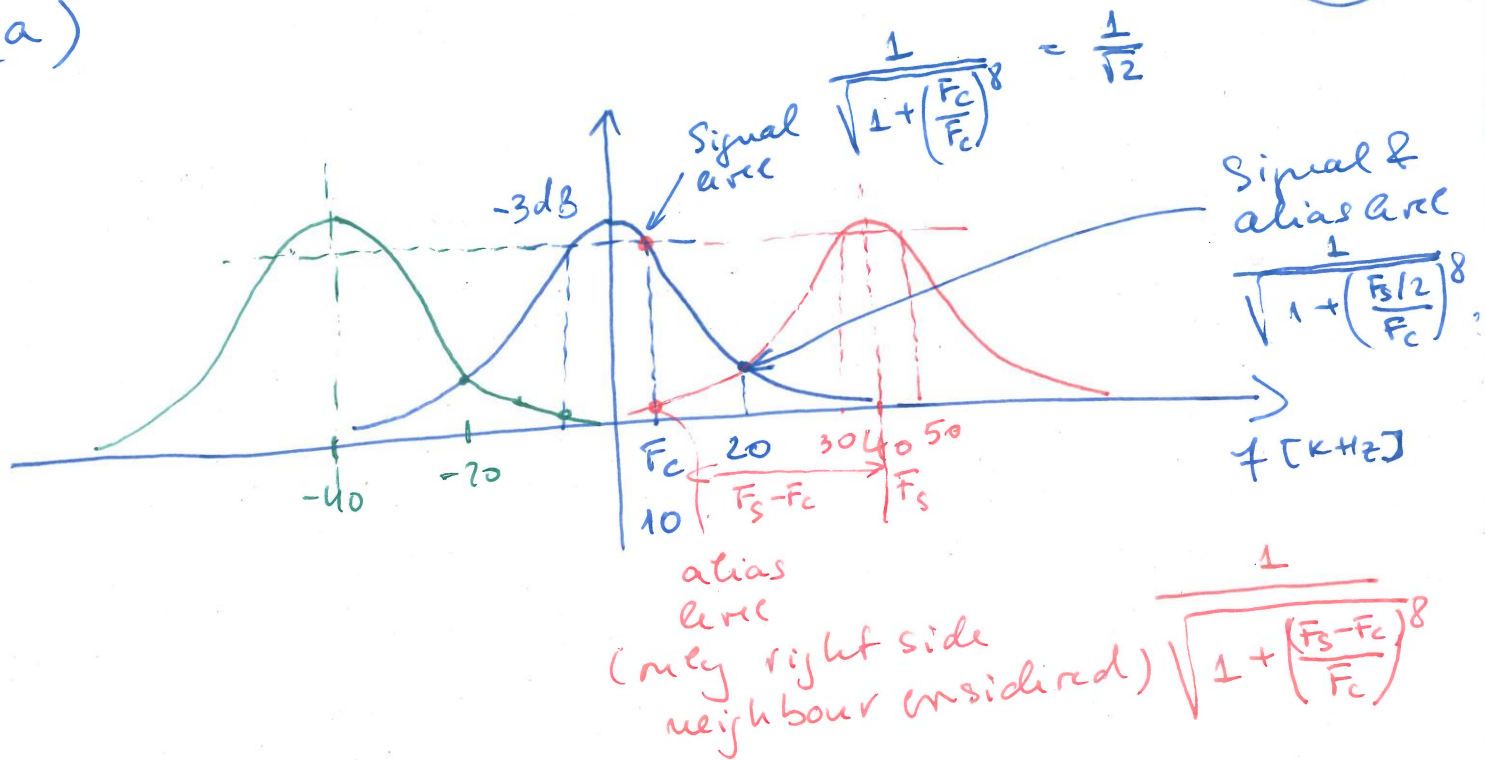
The front-end of a real-time DSP is shown below. Assume a wideband input signal.



- (a) Sketch the spectrum of the signal before sampling (point A) and after sampling (point B) between the range $\pm F_s$.
- (b) Determine the signal and aliasing error levels at 10 kHz and at the Nyquist frequency.
- (c) Determine the minimum sampling frequency $F_s(\text{min})$ to give a signal-to-alias level of 10:1 at 10 kHz.

(a)

5a



Butterworth LP 4th order

$$|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^{2 \times 4}}} = \frac{1}{\sqrt{1 + \left(\frac{f}{f_c}\right)^8}}$$

(b) At 10 kHz = f_c Signal level is $\frac{1}{\sqrt{2}} = 0.707$

Alias level is

$$\frac{1}{\sqrt{1 + \left(\frac{f_s - f_c}{f_c}\right)^8}} = \frac{1}{\sqrt{1 + 3^8}} = \frac{1}{81} = 0.012$$

$$\left(\approx \frac{1}{\sqrt{3^8}} = \frac{1}{3^4} = \frac{1}{81} \right)$$

At 20 kHz = $f_s/2$ $\Rightarrow$ Nyquist freq $\Rightarrow$ equal signal/alias level

$$\frac{1}{\sqrt{1 + \left(\frac{20}{10}\right)^8}} = \frac{1}{\sqrt{1 + 2^8}} \approx \frac{1}{16} = 0.0625$$

(c) At $10\text{kHz} = F_c$

(6a)

Signal level is $\frac{1}{\sqrt{2}} = 0.707$

For 10:1 alias error level must be

$$\frac{0.707}{10} = 0.0707$$

Alias error level is

$$\frac{1}{\sqrt{1 + \left(\frac{F_s - F_c}{F_c}\right)^8}} = 0.0707$$

$$\frac{1}{1 + \left(\frac{F_s - F_c}{F_c}\right)^8} = (0.0707)^2$$

$()^2$

$$1 + \left(\frac{F_s - F_c}{F_c}\right)^8 = \frac{1}{(0.0707)^2}$$

$\sqrt{x}$

-1 $\sqrt[8]{x}$

$$\frac{F_s - F_c}{F_c} = \sqrt[8]{\frac{1}{(0.0707)^2} - 1}$$

$$\frac{F_s}{F_c} - 1 = \sqrt[8]{\frac{1}{(0.0707)^2} - 1}$$

$$\frac{F_s}{F_c} - 1 = 1.938$$

$+1$

$$F_s = F_c \times 2.938 = 29.38\text{kHz}$$

Q3

7

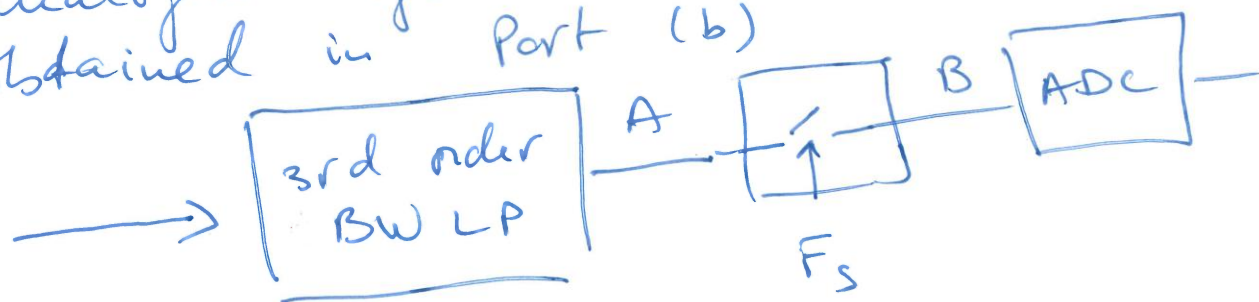
(a) The front-end of a real-time DSP is depicted below. Assume a wideband input signal, $F_c = 20\text{kHz}$ and $F_s = 48\text{kHz}$.

(i) Sketch the signal before (point A) and after sampling (point B) (spectrum of signal)

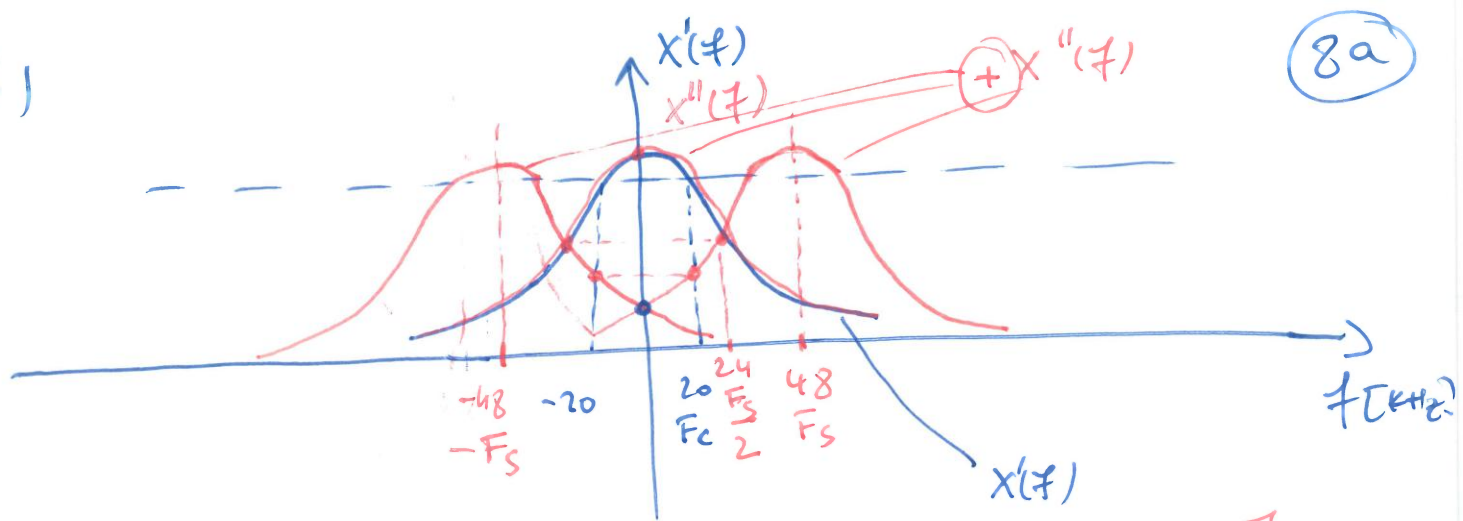
(ii) Determine the signal-to-aliasing error ratio at 0Hz , 10kHz and 20kHz . Comment on the practical implications of your answer.

(b) Determine the minimum sampling frequency $F_s(\text{min})$ to give a signal-to-aliasing error ratio of 50:1 at 20kHz .

(c) What is the aliasing frequency at the cutoff frequency F_c if the analogue signal is sampled at $F_s(\text{min})$ obtained in Part (b)



(i)



8a

Contribution of other aliases with
shown (un-zero in $(-F_s, F_s)$)

(ii) $F = 0 \text{ Hz}$

Signal is 1

$$H(0) = \frac{1}{1} = 1$$

Alias errn is

(only first alias)

$$H(F_s - 0) = \frac{1}{\sqrt{1 + \left(\frac{F_s}{F_c}\right)^6}} =$$

$$= \frac{1}{\sqrt{1 + \left(\frac{48}{20}\right)^6}} = \frac{1}{\sqrt{1 + 191.1}}$$

$$= 0.072$$

$$\text{Signal / alias} = 1 : 0.072 = 13.86 : 1$$

$$F = 10 \text{ kHz}$$

Signal

$$H(F) = \frac{1}{\sqrt{1 + \left(\frac{F}{F_c}\right)^6}} = \frac{1}{\sqrt{1 + \left(\frac{10}{20}\right)^6}} = 0.992$$

Alias error

$$H(F_s - F) = \frac{1}{\sqrt{1 + \left(\frac{F_s - F}{F_c}\right)^6}} = \frac{1}{\sqrt{1 + \left(\frac{48 - 10}{20}\right)^6}}$$

$$= \frac{1}{\sqrt{1 + \left(\frac{38}{20}\right)^6}} = \frac{1}{\sqrt{1 + 47.05}} =$$

$$= 0.1443$$

$$\text{Signal : Alias} = 0.992 : 0.1443$$

$$= 6.88 : 1$$

$$F = 20 \text{ kHz} = F_c$$

Signal

$$H(F_c) = \frac{1}{\sqrt{2}} = 0.707$$

Alias error

$$H(F_s - F_c) = \frac{1}{\sqrt{1 + \left(\frac{F_s - F_c}{F_c}\right)^6}} =$$

$$= \frac{1}{\sqrt{1 + \left(\frac{48 - 20}{20}\right)^6}} = \frac{1}{\sqrt{1 + 7.53}} = 0.342$$

$$\text{Signal / Alias} = 0.707 : 0.342 = 2.06 : 1$$

(b)

(10a)

$$H(20) = H(F_c) = 0.707$$

$$H(F_s - F_c) = \frac{1}{\sqrt{1 + \left(\frac{F_s - F_c}{F_c}\right)^6}} = \frac{0.707}{50} = \frac{1.414}{100}$$

$$= 0.01414$$

$$1 + \left(\frac{F_s - F_c}{F_c}\right)^6 = \frac{1}{(0.01414)^2}$$

$$\left(\frac{F_s}{F_c} - 1\right)^6 = \frac{1}{(0.01414)^2} - 1$$

$$\frac{F_s}{F_c} = \sqrt[6]{\frac{1}{(0.01414)^2} - 1} + 1$$

$$\frac{F_s}{F_c} = 5.135$$

$$\Rightarrow F_s = 5.135 F_c = 5.135 \times 20 \text{ kHz}$$

$$= 102.7 \text{ kHz}$$

(c) F_c is overlapped by

$$F_s - F_c = (102.7 - 20) \text{ kHz}$$

$$= \underline{\underline{82.7 \text{ kHz}}}$$

Q4

11

$$\frac{H(F)}{H(F_s - F)} = \text{Signal / Alias error ratio}$$

$$\frac{1}{\sqrt{1 + \left(\frac{F}{F_c}\right)^{2M}}} \times \sqrt{1 + \left(\frac{F_s - F}{F_c}\right)^{2M}}$$

$$= \sqrt{\frac{1 + \left(\frac{F_s - F}{F_c}\right)^{2M}}{1 + \left(\frac{F}{F_c}\right)^{2M}}} = a$$

$$1 + \left(\frac{F_s - F}{F_c}\right)^{2M} = a^2 \left(1 + \left(\frac{F}{F_c}\right)^{2M}\right)$$

$$1 + \left(\frac{F_s - F}{F_c}\right)^{2M} = a^2 + a^2 \left(\frac{F}{F_c}\right)^{2M}$$

$$1 - a^2 = a^2 \left(\frac{F}{F_c}\right)^{2M} - \left(\frac{F_s - F}{F_c}\right)^{2M}$$

$$1 - a^2 = \frac{1}{\left(\frac{F_c}{F}\right)^{2M}} \left(a^2 F^{2M} - (F_s - F)^{2M} \right)$$

$$\bar{F}_c^{2n} = \frac{a^2 \bar{F}^{2n} - (\bar{F}_s - \bar{F})^{2n}}{1 - a^2}$$

$$\bar{F}_c = \sqrt[2n]{\frac{(\bar{F}_s - \bar{F})^{2n} - a^2 \bar{F}^{2n}}{-1 + a^2}}$$

$$\boxed{\bar{F} = 0}$$

$$\bar{F}_c = \frac{\sqrt[2n]{\bar{F}_s^{2n}}}{\sqrt[2n]{a^2 - 1}} = \frac{\bar{F}_s}{\sqrt[2n]{a^2 - 1}}$$

RC n=1

$$\bar{F}_c = \frac{\bar{F}_s}{\sqrt{a^2 - 1}} \approx \frac{\bar{F}_s}{a} \quad a \gg 1$$

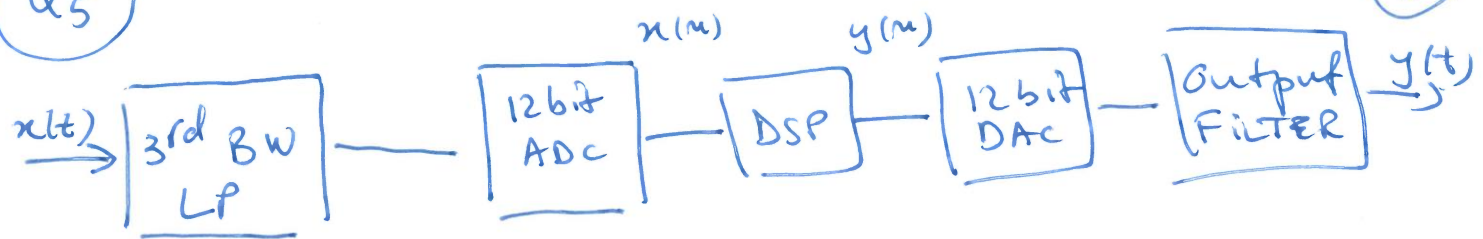
$$\begin{aligned} \text{If } a = 10 &\Rightarrow \bar{F}_c \approx \frac{\bar{F}_s}{10} = 2500 \text{ Hz} \\ a = 20 &\Rightarrow \bar{F}_c \approx \frac{\bar{F}_s}{20} = 1250 \text{ Hz} \end{aligned}$$

Sallen-Key n=2

$$\bar{F}_c = \frac{\bar{F}_s}{\sqrt[4]{a^2 - 1}} \approx \frac{\bar{F}_s}{\sqrt{a}} \quad a \gg 1$$

$$a = 10 \Rightarrow \bar{F}_c \approx \frac{\bar{F}_s}{\sqrt{10}}$$

Q5



13

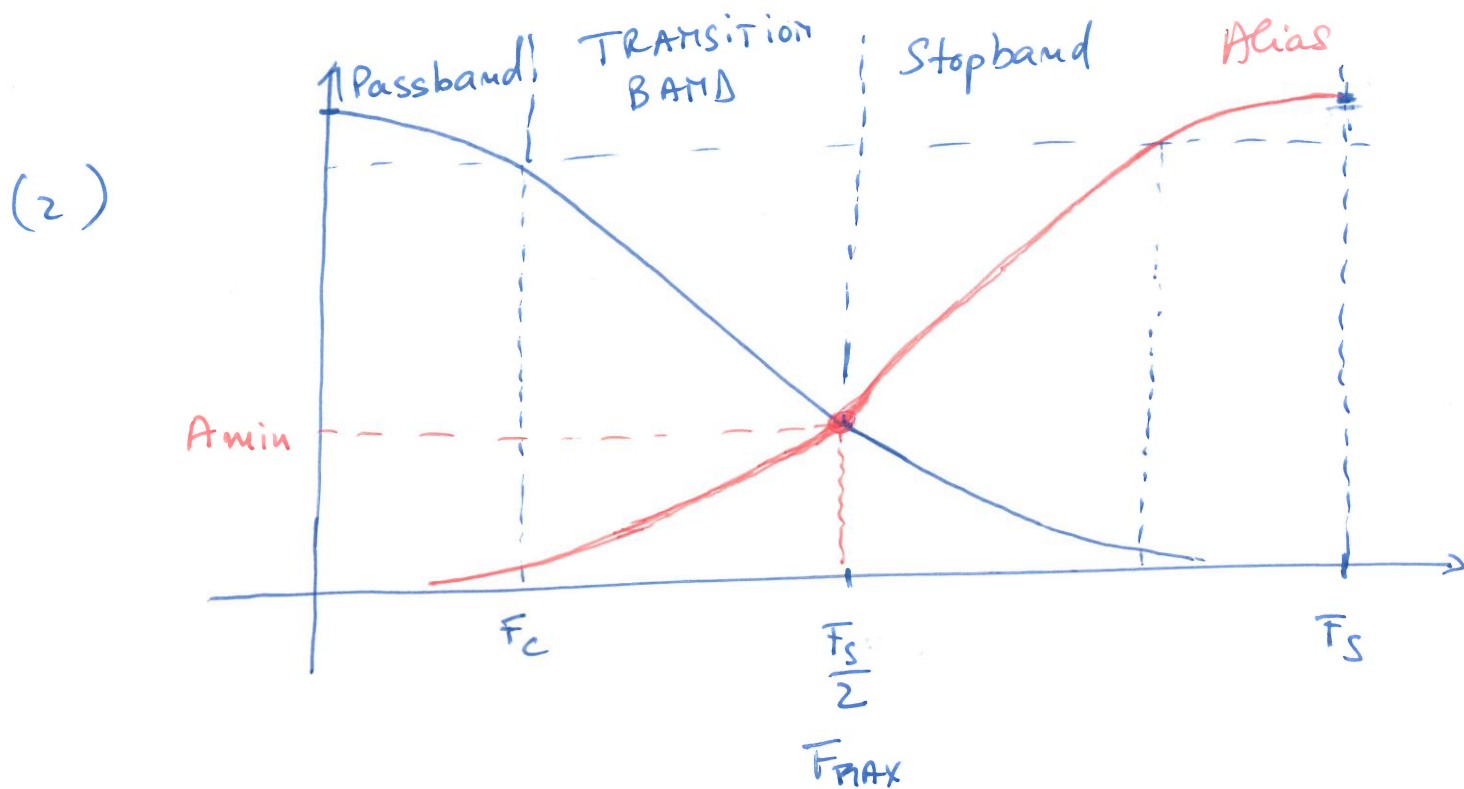
Passband 0-4 kHz

Dipolar ADC

- (1) Minimum stopband attenuation for the anti-aliasing filter A_{min}
- (2) minimum sampling frequency F_s
- (3) Signal to aliasing error level for A_{min} and F_s in the passband

$$\begin{aligned}
 (1) \quad \text{SQNR} &= 6.02B + 1.76 \text{ dB} \\
 &= 20 \log_{10} (\sqrt{1.5} \times 2^B) \\
 &= 10 \log_{10} (1.5 \times 2^{2B})
 \end{aligned}$$

$$\text{SQNR} = 6.02 \times 12 + 1.76 = 72.48 + 1.76 \text{ dB} \approx 74 \text{ dB}$$



$$20 \log_{10} \frac{1}{\text{HH}} = 20 \log_{10} \sqrt{1 + \left(\frac{F_{\text{MAX}}}{F_c} \right)^6} = A_{\text{min}}$$

$$\Rightarrow 20 \log_{10} \sqrt{1 + \left(\frac{F_{\text{MAX}}}{F_c} \right)^6} = 74$$

$$\sqrt{1 + \left(\frac{F_{\text{MAX}}}{F_c} \right)^6} = 10^{\frac{74}{20}} = 5011.87$$

$$\left(\frac{F_{\text{MAX}}}{F_c} \right)^6 = (5011.87)^2 - 1$$

$$\begin{aligned}
 F_{\text{MAX}} &= F_c \times \sqrt[6]{(5011.87)^2 - 1} = 17.113 \times F_c \\
 &= 68.45 \text{ kHz}
 \end{aligned}$$

4 kHz
||

(3)

15a

$$F_S = 2F_{MAX} = 136.9 \text{ kHz}$$

Signal at F_c $\frac{1}{\sqrt{2}} = 0.707$

Alias error at F_c

$$\frac{1}{\sqrt{1 + \left(\frac{F_S - F_c}{F_c}\right)^6}} = \frac{1}{\sqrt{1 + \left(\frac{136.9 - 4}{4}\right)^6}} =$$

$$= \frac{1}{\sqrt{1 + \left(\frac{132.9}{4}\right)^6}}$$

$$= \frac{1}{\sqrt{1 + (33.2)^6}}$$

$$= 0.0000273$$

$$= 2.73 \times 10^{-5}$$

$$\text{Signal / Alias} = \frac{0.7071}{2.73 \times 10^{-5}} = 25872.2 : 1$$

large

Q6

16

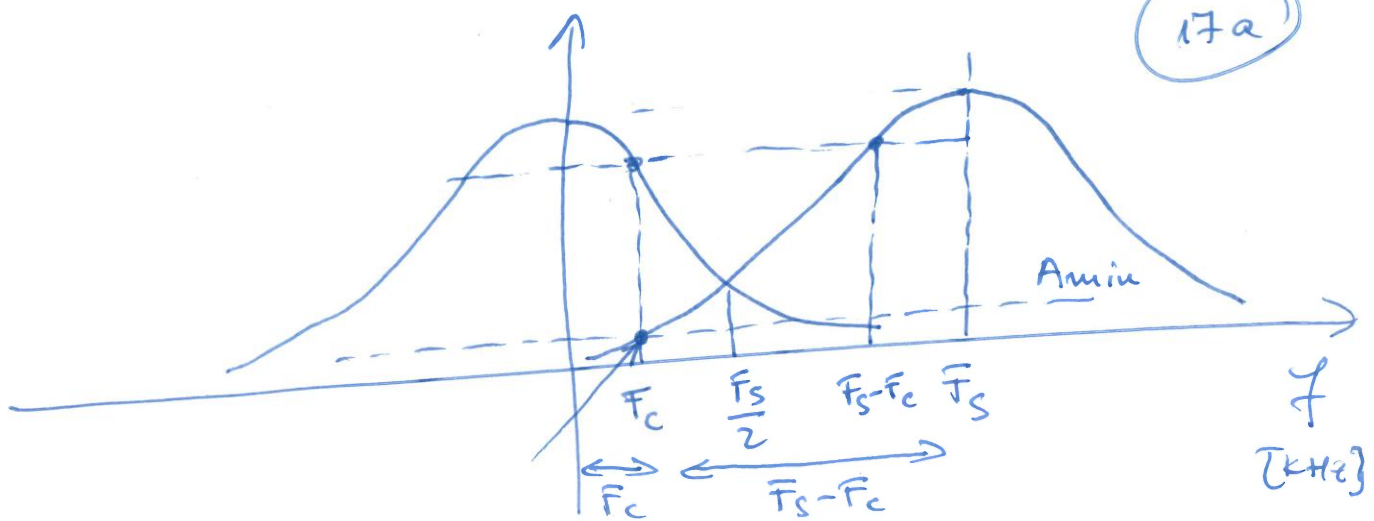
An analogue signal with uniform power spectrum density is bandlimited by an anti-aliasing filter with the following amplitude response:

$$|H(f)| = \frac{1}{\sqrt{1 + \left(\frac{f}{F_c}\right)^8}}$$

Where $F_c = 5 \text{ kHz}$. The signal is digitised with a linear 12-bit bipolar ADC

(1) The minimum sampling frequency to keep the maximum aliasing error in the passband to no greater than the quantization error level.

(2) The maximum passband signal level in dB relative to the ADC quantization noise floor



Aliasing error at this point needs
to be $\leq A_{\min}$

$$20 \log_{10} \left| \frac{1}{H(f)} \right|^2 = A_{\min} = 6.02 B + 1.76 \text{ dB}$$

$$B = 12$$

$$\Rightarrow A_{\min} = 74 \text{ dB}$$

$$\Rightarrow \left| \frac{1}{H(f)} \right|^2 = 10^{\frac{74}{10}}$$

$$f = F_s - F_c$$

$$1 + \left(\frac{F_s - F_c}{F_c} \right)^8 = 10^{\frac{74}{10}}$$

$$\left(\frac{F_s - F_c}{F_c} \right)^8 = 10^{\frac{74}{10}} - 1$$

$$\frac{F_s}{F_c} - 1 = \sqrt[8]{10^{\frac{74}{10}} - 1}$$

$$= 8.414$$

$$\frac{F_s}{F_c} = 9.414$$

$$F_s = 9.414 \times F_c = \underline{\underline{47 \text{ kHz}}}$$

Note The notes have an error as $n=3$ was used in the solution instead of $n=4$

This gives

$$\frac{F_s}{F_c} - 1 = \sqrt[6]{10^{\frac{74}{10}} - 1} = 17.11$$

$$F_s = 18.11 \times F_c = 90.57 \text{ kHz}$$

(b) It is $A_{\min} = 74 \text{ dB}$